

Introduction to Database Systems

Homework 4

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Fall 2024

Do as many of the following exercises as you have time for. Focus on the exercises you feel will benefit your preparation the most. To pass the homework, you need to have *15/80* points.

1 Hardware and Transactions [10 points]

Question 1.A Select the correct statements below:

- (a) NO STEAL is an optimal Buffer Management policy.
- (b) In a main memory system, it is not necessary to include the “old” values when logging changes made by transactions to secondary storage.
- (c) With 2-Phase Locking, no locks can be acquired once the shrinking phase has begun.
- (d) In a distributed system, it is possible to avoid network partitions and achieve consistency and availability.

Question 1.B Reflect upon why ACID transactions are rarely used in distributed systems.

2 Data Systems for Analytics [10 points]

Question 2.A Select the correct statements below:

- (a) Graph databases always perform better than relational databases for graph-specific workloads.
- (b) MapReduce systems such as Apache Hadoop scale well horizontally by distributing data across multiple servers and performing computations in parallel on these servers.

- (c) The concept of “veracity” in big data refers to the accuracy and reliability of the data being analyzed.
- (d) It is easy to manage semi-structured data in relational database management systems.
- (e) Relational database systems and SQL are well-suited for machine learning applications that involve many loops.
- (f) Big data analytics applications can only be implemented on systems that support ACID transactions.

Question 2.B Consider a scenario where you have 100 PB of raw data files generated from a scientific experiment. Your goal is to make scientific discoveries by performing complex computations on this data and to plan future experiments accordingly. Provide compelling arguments for why you would choose to store this data in a data lake and use a data lakehouse for processing. Compare this to a scenario in which the data is stored on a file system in the research group’s local data center and additionally loaded into a document database installed on the cloud for more effective data analysis.

3 Normalisation [20 points]

Consider the SQL script and the associated description in Part 2 of Homework 3. In Homework 3, you normalised the Rentals relation. Now, your task is to normalise the Projects relation. For that purpose, follow the steps outlined below:

1. Find all the FDs in the relations, given the constraints and assumptions from Homework 3.
2. Decompose the relation until each subrelation is in BCNF/3NF, while preserving all non-redundant FDs. Write down the resulting schema description in a simple Relation(columns) format.
3. Write the detailed SQL commands to create the resulting tables (with primary keys and foreign keys) and populate them, by extracting the relevant data from the original relations.
4. Select the correct normal form for the decomposed schema.

Please note that the details of these steps, as well as constraints of the relation, can be found in the description of Homework 3. Please refer to the description there for further guidance.

4 Godly strife (by Johan von Tangen Sivertsen) [20 points]

Zeus is tired of all the infighting among the gods. He has brought you to Mount Olympus to help deal with the problem. To avoid conflicts, Zeus wants a simple database that contains a description of each promise that the gods have made to humans. That way, a god can quickly check for conflicts before making a new promise. In case conflicts arise anyway, the database should track the various sacrifices made by humans to the gods. The time, place, and items sacrificed should all be recorded. Sacrifices are divisible into one of three categories, flesh, wine, and valuables (e.g., gold, gemstones, etc.). Each category has different attributes that describe them, but they all have a value. It should be possible to sum up the total value of sacrifices a human has offered to a God to help resolve any conflicting promises.

Furthermore, some humans are appointed priests. A priest is always promised protection by his associated God, and if a sacrifice is conducted in a ceremony presided over by a priest, it is twice as valuable as normally. Any priest serves a single God. Finally, Poseidon has requested that the database also tracks if a human attacks any sea monster or cyclops, and in that case, a hecatomb should be deducted from their total sacrifice value for all gods.

1. Draw an ER diagram that supports the requirements. Feel free to add identifiers and other attributes.
2. Write the DDL for a database according to your design.

Remember that Zeus will likely strike you down with lightning if he is not pleased with your work!

5 SQL [20 points]

Note: The following queries are relatively simple. In the exam, there will also be some more advanced queries, including division queries. Please, go through the old exams for some examples of more challenging queries.

In this part you will work with a music database. To start working with the database, import/run `HW4.sql` found in LearnIT using the PostgreSQL DBMS on your laptop. The database has the following relations:

```
Artists(ArtistId, Artist, ArtistImageUrl)
Songs(SongId, Title, ArtistId, Duration, IsExplicit, ImageUrl, ReleaseDate)
Genres(GenreId, Genre)
Albums(AlbumId, Album, AlbumImageUrl, AlbumReleaseDate)
```

```
AlbumArtists(AlbumId, ArtistId)
AlbumGenres(AlbumId, GenreId)
AlbumSongs(AlbumId, SongId)
SongGenres(SongId, GenreId)
```

Primary and foreign key attributes have names that end with `Id`. The meaning of other attributes should be self-explanatory. The first four relations have their first attribute as primary key. In the last four relations there is a composite primary key consisting of both attributes, and each attribute separately is a foreign key reference to one of the first four relations. Secondary indexes exist on `AlbumArtists(ArtistId)`, `AlbumGenres(GenreId)`, `AlbumSongs(SongId)`, `SongGenres(GenreId)`, and `Songs(ArtistId)`.

You will need to work with the `ReleaseDate` and `Duration` attributes, which have the type `date` and `time`, respectively. The expression `interval '1 minute'` can be used to generate a duration value to compare to, the expression `extract(year from ReleaseDate)` can be used to get a year from a date, and `extract(epoch from Duration)` can be used to get the total number of seconds in an interval.¹

Answer each of the following questions using a single SQL query on the examination database. Enter the result of each query into the quiz on LearnIT. As before, queries should adhere to the detailed guidelines given in Homework 1.

- (a) In the database, 353 songs have a duration of at least 10 minutes. What is the *average* duration of songs, in minutes, that have a duration between 5 and 25 minutes, inclusive? Round the number of minutes (`ROUND(...)`).
- (b) What is the total duration *in minutes* of all explicit songs in the database? Round the number of minutes (`ROUND(...)`).
- (c) The database contains just 5 songs released in 1953. What is the *average* number of songs released in a year? Round the number of songs (`ROUND(...)`).

Note: This is a very simple query. Try also to answer which year had the largest number of songs. Observe how much harder this query is!

- (d) The database contains multiple albums by the artist **Queen**. Each album has a different average song duration, with the maximum average song duration of an album by **Queen** being 354 seconds. What is the maximum average song duration (in seconds) of an album by **Miles Davis**?

Note: The output of the maximum average song duration is rounded `ROUND(...)`

- (e) There are 938 song titles that have been used for at least 2 songs, making up a total of 2072 songs with those titles. How many songs have a title that has been used for at least 4 songs?

¹For more details, see: <https://www.postgresql.org/docs/current/functions-datetime.html>.

- (f) How many songs have been released after 2010 or belong to an album released in January.
- (g) There are 1147 Albums with more than 1 song and none of them are Explicit. How many Albums consists of more than 1 song with all songs being Explicit?
- (h) The highest number of genres covered within an Album is 5. In the database, there is only one Album that has this amount of genres. What is the name of this Album?

Note: Write your query to be capable of finding all albums that have the highest number of genres. (No hardcoded values)

HW4 - Example Solution

(1) 1.A) Hardware and DBMS Design

MULTIPLE CHOICE marked out of 5 penalty 0.10 Multiple answers allowed

1.A) Select the true statements:

- a. NO STEAL is an optimal Buffer Management policy. (−50%)
- b. In a main memory system, it is not necessary to include the “old” values when logging changes made by transactions to secondary storage. (50%)
- c. With 2-Phase Locking, no locks can be acquired once the shrinking phase has begun. (50%)
- d. In a distributed system, it is possible to avoid network partitions and achieve consistency and availability. (−50%)

(2) 1.B) Hardware and DBMS Design

ESSAY marked out of 5 penalty 0.10 Plain text, monospaced font

1B) Write your reflections here:

Information for graders:

- In the context of ACID transactions, consistency refers primarily to maintenance of primary keys and foreign keys. While globally unique identifiers are feasible, using foreign keys would require either the ability to partition the data such that those constraints can be checked locally, or significant work to check constraints at update time. Since data collections that are large enough to require distributed systems are rarely relational, this effort is not considered worthwhile.

(3) 2.A) Data Systems for Analytics

MULTIPLE CHOICE marked out of 5 penalty 0.10 Multiple answers allowed

2.A) Select the true statements:

- a. Graph databases always perform better than relational databases for graph-specific workloads. (−33.33333%)
- b. MapReduce systems such as Apache Hadoop scale well horizontally by distributing data across multiple servers and performing computations in parallel on these servers. (33.33333%)
- c. The concept of “veracity” in big data refers to the accuracy and reliability of the data being analyzed. (33.33333%)
- d. It is easy to manage semi-structured data in relational database management systems. (33.33333%)
- e. Relational database systems and SQL are well-suited for machine learning applications that involve many loops. (−33.33333%)

- f. Big data analytics applications can only be implemented on systems that support ACID transactions. (−33.33333%)

(4) **2.B) Data Systems for Analytics**

ESSAY marked out of 5 penalty 0.10 Plain text, monospaced font

2.B) Write your arguments here: Data duplication in the second scenario is problematic, as it creates uncertainty about the source of truth when data is modified either in the file system or the cloud database. Additionally, loading data into the cloud database is time-consuming and expensive. In contrast, data lakes, which use open data formats, offer several advantages: they support a wide range of tools, facilitate easier data sharing among researchers from different groups or universities, and help avoid duplication issues.

Information for graders:

(5) **3.1 (Projects) FDs**

ESSAY marked out of 5 penalty 0.10 Plain text, monospaced font

3.1 (Projects) Write down FDs for the Projects relation.

Information for graders:

- $PID \rightarrow PN, SID \rightarrow SN, ID \rightarrow MID, MID \rightarrow MN$

(6) **3.2 (Projects) Decomposition**

ESSAY marked out of 5 penalty 0.10 Plain text, monospaced font

3.2 (Projects) Write down the decomposed relations for the Projects relation.

Information for graders:

- Projects1(MID, MN)
Projects2(ID, MID)
Projects3(PID, PN)
Projects4(SID, SN)
Projects0(ID, PID, SID)

(7) **3.3 (Projects) DDL**

ESSAY marked out of 5 penalty 0.10 Plain text, monospaced font

3.3 (Projects) Write down SQL commands to create and populate the decomposed relations for Projects.

Information for graders:

- See Homework4-SOL3.sql

(8) **3.4 (Projects) Normal Form**

MULTIPLE CHOICE marked out of 5 penalty 0.10 One answer only

3.4 (Projects) Select the lowest normal form for any relations of the Projects schema:

- a. All relations are in BCNF (and therefore also 3NF) ✓
- b. Some relations are only in 3NF, not in BCNF

(9) **4.1) ER-diagram**

ESSAY marked out of 10 penalty 0.10 Plain text, monospaced font

4.1) Submit ER-diagram.

Information for graders:

- See Homework4-SOL4.PDF

(10) **4.2) DDL**

ESSAY marked out of 10 penalty 0.10 Plain text, monospaced font

4.2) SQL DDL to implement the diagram.

Information for graders:

- See Homework4-SOL4.sql

(11) **5(a) Numerical answer**

NUMERICAL marked out of 2.5 penalty 0.10

5(a) Run the query and paste the result here (an integer):

- 7 (0%)

(12) **5(b) Numerical answer**

NUMERICAL marked out of 2.5 penalty 0.10

5(b) Run the query and paste the result here (an integer):

- 1674 ✓

(13) **5(c) Numerical answer**

NUMERICAL marked out of 2.5 penalty 0.10

5(c) Run the query and paste the result here (an integer):

- 240 ✓

(14) **5(d) Numerical answer**

NUMERICAL marked out of 2.5 penalty 0.10

5(d) Run the query and paste the result here (an integer):

- 578 ✓

(15) **5(e) Numerical answer**

NUMERICAL marked out of 2.5 penalty 0.10

5(e) Run the query and paste the result here (an integer):

- 132 ✓

(16) **5(f) Numerical answer**

NUMERICAL marked out of 2.5 penalty 0.10

5(f) Run the query and paste the result here (an integer):

- 9682 ✓

(17) **5(g) Numerical answer**

NUMERICAL marked out of 2.5 penalty 0.10

5(g) Run the query and paste the result here (a floating point number):

- 14 ✓

(18) **5(h) Textual answer**

NUMERICAL marked out of 2.5 penalty 0.10

5(h) Run the query and paste the result here (an integer):

- Krywan, Krywan ✓

Total of marks: 80